



Building and Evaluating a Mini Web Search Engine with Cross-Lingual Retrieval

1. Problem Statement

Design and implement a mini web search engine capable of retrieving relevant documents from a domain-specific corpus (choose any domain of your choice). The system must support:

- Ranked retrieval using the Vector Space Model
- Text mining enhancements
- Cross-lingual retrieval (Example: English ↔ Hindi)
- Rigorous evaluation using standard IR metrics

2. Objectives

1. Build an end-to-end retrieval pipeline
2. Implement and analyse Vector Space Model
3. Integrate at least one text mining technique
4. Support cross-lingual query processing
5. Conduct systematic evaluation

3. Dataset Requirements

You may use datasets openly available fitting your domain having at least 20% documents in any language other than English.

4. Technical Requirements

4.1 Preprocessing

- Tokenization
- Stop-word removal
- Stemming or lemmatization
- Construction of inverted index



4.2 Vector Space Model (Mandatory)

- TF and TF-IDF weighting
- Cosine similarity

4.3 Text Mining Component (Choose at least one)

- Topic modelling
- Document clustering
- Query expansion
- Keyphrase extraction

Explain how it improves retrieval.

4.4 Cross-Lingual Retrieval (Mandatory)

Implement one of:

- Query translation approach
- Shared embedding approach
- Dictionary-based mapping

System must support (For English - Hindi):

- Hindi query → English documents
- English query → Hindi documents

4.5 Web Search Features

- Ranked top-10 retrieval
- Snippet generation
- Highlighted query terms

UI can be minimal (CLI/Streamlit/Flask).

5. Evaluation Requirements

- Create 20–30 evaluation queries
- Compute:
 - Precision
 - Recall
 - MAP



6. Deliverables

1. A zip file containing source code, technical report and assignment report strictly in PDF format including Visualization of clusters.
2. Naming convention for assignment, Group#--assignment-2.zip
3. Whatever assumptions you are making mention it clearly and why
4. Strictly adhere to assignment guidelines, and there will be no extension under any circumstances
5. Make sure to submit assignment as PDF file only
6. If zip file size including code is more than 10 MB, share source code link in pdf file.